Instituto Superior Técnico

BSc Aerospace Engineering

Multivariable, Nonlinear and Optimal Control

Final Report: Linear State Feedback Control of an Inverted Pendulum

Professor: João Manuel Lage de Miranda Lemos

Laboratory Professor: Carlos Filipe Gomes Bispo

Grup: 35

Lucas Moreira Nº 109657

Tomás Teixeira Nº 109766

Pedro Machado Nº 110718

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I. Introdução

A. Context and Objectives

This report details the work developed within the scope of the Multivariable, Non-linear, and Optimal Control course laboratory project. The primary objective is to design, simulate, and implement a robust control strategy to stabilize a highly non-linear and inherently unstable system using a pre-existing experimental setup of a rotary inverted pendulum.

The adopted methodology relies on linearizing the plant's mathematical model around its unstable equilibrium point (the upward vertical position) to derive a Linear Time-Invariant (LTI) state-space representation. Based on this model, a Linear Quadratic Gaussian (LQG) control architecture is developed. This comprises a Linear Quadratic Regulator (LQR) to compute the optimal state feedback law, and a Linear Quadratic Estimator (LQE, or Kalman Filter) to estimate the system's full internal state from noisy measurements. The project covers the theoretical analysis of the system, validation through MATLAB/Simulink simulations, and practical real-time implementation on the laboratory's physical hardware, specifically addressing real-world non-linearities such as motor deadzones and steady-state errors.

B. System Description

The physical plant utilized in this work is a rotary inverted pendulum apparatus (a Quanser experimental setup). The mechanical structure consists of a motorized rotary arm that actively maneuvers an unactuated, free-swinging pendulum rod attached to its extremity.

The system operates as a Single-Input Multiple-Output (SIMO) process. The single control input u is the continuous voltage applied to the DC motor that drives the horizontal rotary arm. To monitor the system's behavior, two physical sensors provide the measured output vector y:

- A potentiometer measuring the angular position of the rotary arm α.
- An optical encoder measuring the precise angular position of the pendulum relative to the arm β.

Because the system's complete state vector includes both the angular positions α, β and their respective angular velocities $\dot{\alpha}, \dot{\beta}$, but only the positions are directly measured, the system requires a state observer. Furthermore, the provided physical setup interfaces with the Simulink control environment via A/D and D/A data acquisition boards, allowing the designed controllers to be executed and tested on the real system in real-time.

II. Theoretical Analysis and Control Design

A. Space-State Formulation

A dinâmica não-linear do distema implica que este necessite de ser linearizado em torno de um ponto de equilíbrio, que dado o objetivo deste relatório, é o ponto de equilíbrio instável onde o pêndulo se encontra na posição vertical para cima. Este equilíbrio corresponde a todas as variáveis de estado iguais a zero. A derivação do modelo não-linear e a subsequente linearização não são objeto deste relatório, podendo ser encontrados no Capítulo 2 da tese de Samuel Balula **(fazer referência)**. O modelo linearizado do sistema que representa a dinâmica razoavelmente bem é dado por:

$$\begin{cases} \dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \\ \mathbf{y} = C\mathbf{x} + D\mathbf{u} \end{cases}$$

onde $\mathbf{x} = [\alpha\ \dot{\alpha}\ \beta\ \dot{\beta}\ i]^T$ é o vetor de estados de dimensão 5 e i é a corrente do motor. O vetor de entrada unidimensional $\mathbf{u} = u$ corresponde à diferença de potencial contínua, em *Volts*, aplicada nos terminais do motor DC através de um power amplifier, controlando diretamente α. O vetor de saída $\mathbf{y} = [\alpha\ \beta]^T$ é composto pelas medições dos sensores, correspondendo às posições angulares do braço e do pêndulo, respetivamente.

As matrizes do modelo A, B, C e D

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & -0.0174 & 20.7861 & -0.0023 & 57.5344 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -0.0174 & 63.8319 & -0.0071 & 57.4388 \\ 0 & -232.0252 & 0 & 0 & -755.425 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 333.3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

B. State-Space Analysis

In this section, we analyze the previously formulated state-space model in terms of system poles, controllability, observability, and frequency response.

1) Q1 - System Poles: The system poles are the eigenvalues of the dynamics matrix A. Using MATLAB's `eig()` function, the following poles were obtained:

Tabela I: Características dos Modos do Sistema

Polos	ω (rad/s)	τ (s)
0	0	∞
-737.3	7.37×10^2	1.36×10^{-3}
-19.36	1.94×10^1	5.16×10^{-2}
-5.761	5.76	1.74×10^{-1}
6.997	7.00	-1.43×10^{-1}

Analysis of the poles shows that the system has three stable modes: two correspond to the mechanical damping of the pendulum, $\lambda_3 = -19.3673$ and $\lambda_4 = -5.7612$, and a very fast mode associated with the motor current, $\lambda_2 = -737.3184$. Additionally, the state α generates a pole at the origin, $\lambda_1 = 0$, due to being an integrator. This implies that after a perturbation, α will accumulate error over time without diverging or converging, reflecting the fact that the arm rotates without friction and does not naturally settle to an equilibrium.

The single unstable mode, $\lambda_5 = 6.9974$, is related to the vertical pendulum position, which is the primary state we aim to control.

2) Q2 - Controllability: A system is controllable if it is possible to transition from any state x_1 to another arbitrary state x_2 by applying an input $u(t)$ over a finite time interval. Knowing whether the system is controllable is crucial for understanding the limitations of the controller.

The controllability matrix is defined as:

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix}, \tag{1}$$

where n is the number of states. Using MATLAB's `ctrb(A,B)` function, we obtained:

$$\mathcal{C} = \begin{bmatrix} 0 & 0 & 1.918 \times 10^4 & -1.449 \times 10^7 & 1.069 \times 10^{10} \\ 0 & 1.918 \times 10^4 & -1.449 \times 10^7 & 1.069 \times 10^{10} & -7.881 \times 10^{12} \\ 0 & 0 & 1.915 \times 10^4 & -1.446 \times 10^7 & 1.067 \times 10^{10} \\ 0 & 1.915 \times 10^4 & -1.446 \times 10^7 & 1.067 \times 10^{10} & -7.869 \times 10^{12} \\ 3.333 \times 10^2 & -2.518 \times 10^5 & 1.858 \times 10^8 & -1.370 \times 10^{11} & 1.010 \times 10^{14} \end{bmatrix}$$

The system is controllable if the rank of the controllability matrix equals the number of states. Using MATLAB's `rank()` function, we obtained $\text{rank}(\mathcal{C}) = 5$, which equals the number of states. Therefore, the system is controllable.

3) Q3 - Observability: A system is observable if, for any initial state $x(0)$, knowledge of the input $u(t)$ and output $y(t)$ over a finite time interval uniquely determines $x(0)$. Although the system is controllable, implementing a state-feedback controller requires knowledge of the full state[1]. Since the state is not directly measurable, it must be reconstructed from the outputs, which is only possible if the system is observable.

The observability matrix is defined as:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \tag{2}$$

Using MATLAB's `obsv()` function, the observability matrix was calculated as:

$$\mathcal{O} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -1.740 \times 10^{-2} & 2.079 \times 10^1 & -2.316 \times 10^{-3} & 5.753 \times 10^1 \\ 0 & -1.738 \times 10^{-2} & 6.383 \times 10^1 & -7.114 \times 10^{-3} & 5.744 \times 10^1 \\ 0 & -1.335 \times 10^4 & -5.096 \times 10^{-1} & 2.079 \times 10^1 & -4.346 \times 10^4 \\ 0 & -1.333 \times 10^4 & -8.152 \times 10^{-1} & 6.383 \times 10^1 & -4.339 \times 10^4 \\ 0 & 1.008 \times 10^7 & -2.762 \times 10^5 & 3.027 \times 10^1 & 3.207 \times 10^7 \\ 0 & 1.007 \times 10^7 & -2.729 \times 10^5 & 2.960 \times 10^1 & 3.202 \times 10^7 \end{bmatrix} \tag{3}$$

Applying the `rank()` function shows that $\text{rank}(\mathcal{O}) = 5$, which equals the number of states. Therefore, the system is observable.

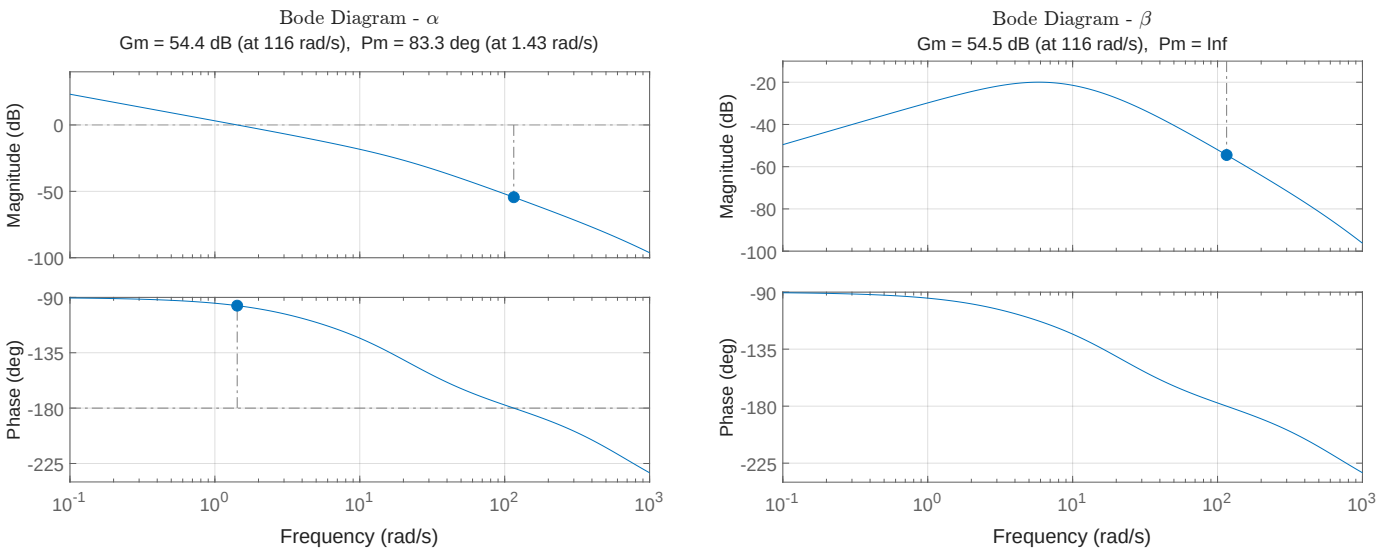
4) Q4 - Bode Plots: The evaluate the frequency response of the system, the transfer functions of α and β relative to the input u were determined, resorting to the function `ss2tf()`:

$$\frac{\alpha(s)}{u(s)} = \frac{1.92 \times 10^4\, s^2 + 92.07\, s - 8.26 \times 10^5}{s^5 + 755.45\, s^4 + 1.33 \times 10^4\, s^3 - 4.82 \times 10^4\, s^2 - 5.76 \times 10^5\, s} \tag{4}$$

$$\frac{\beta(s)}{u(s)} = \frac{0.1915\, s^2}{s^5 + 755.45\, s^4 + 1.33 \times 10^4\, s^3 - 4.82 \times 10^4\, s^2 - 5.76 \times 10^5\, s} \tag{5}$$

With the above transfer functions, we generated the bode plots the caracterize their frequency response.

[1]State-feedback control requires knowledge of the full state since the control law $u(t) = -Kx(t)$ explicitly depends on the state vector. In this project, an estimator was designed to reconstruct the state, resulting in the control law $u(t) = -K\hat{x}(t)$. State estimation is only possible if the system is observable, making observability a prerequisite for implementing state-feedback control when the state is not directly measurable.



(a) Bode plot for $\frac{\alpha(s)}{u(s)}$ (b) Bode plot for $\frac{\beta(s)}{u(s)}$

Figura 1: Open-loop Bode plots of the system evaluating both outputs.

The frequency response of $\frac{\alpha(s)}{u(s)}$ is characterized by a large gain margin, $G_m = 54.4\,dB$ and a large phase margin $P_m = 83.3°$. For frequencies below 1 rad/s, $\frac{\alpha(s)}{u(s)}$ amplifies inputs, whereas for higher frequencies this transfer functions attenuates inputs. The transfer function $\frac{\beta(s)}{u(s)}$ is characterized by a large gain margin, $G_m = 54.4\,dB$, and an infinite phase margin, because the gain never reaches zero.

C. LQR Controller Design

The initial approach to controlling the system involved designing a simple Linear Quadratic Regulator, implementing the control law $u(t) = -Kx(t)$. The associated space-state model is $\dot{x} = A - BK$ and since the pair (A, B) is controllable, it is possible to place the poles of the system in any arbitrary position. To calculate the optimal gain, K, it is necessary to define a cost function J:

$$J = \frac{1}{2}\int_0^\infty \mathbf{x}^T Q\mathbf{x} + \mathbf{u}^T R\mathbf{u}\ dt \tag{6}$$

Where Q_r is the process covariance matrix and R_r is the measurement covariance matrix. The ratio of Q_r to R_r indicates how much the system trust the measurements as opposed to the model predictions. Generally, higher values in Q_r paired with low values of R_r indicate that the system trusts measurements more. Based on the associated space-state model, we designed a preliminar simulink to simulate the model.

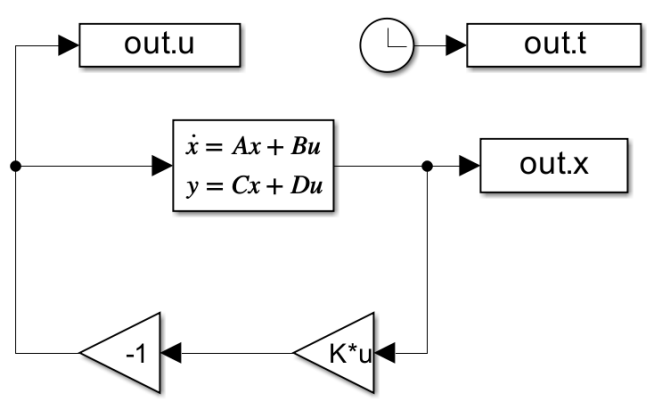


Figura 2: Simulink of the LQR of the system

D. State Estimator Design - LQE

Because the observed output is only α and β, it is necessary to design an estimator that computes the entire state vector from the observed states. Since the system is observable, it is guaranteed that the state vector can always be reconstructed by observing state α and β over an interval of time.

E. Baseline LQG Simulation

Q8

III. Laboratory Work and Experimental Results

A. Initial Hardware Testing

1) LQE Validation: Before testing the performance of the LQG controller, the performance of the LQE was strictly validated on the physical hardware. The primary objective was to ensure that the estimated state vector, $\hat{x}$, accurately tracked the physical system's dynamics—specifically the angular positions α and β

A fundamental observation regarding the experimental setup was the high precision and discrete nature of the sensors. The Quanser platform utilizes a potentiometer for the rotary arm and a high-resolution optical encoder for the pendulum angle. Due to their discrete characteristics and high accuracy, the measurement noise was found to be negligible, effectively implying a standard deviation of zero in ideal conditions. However, to maintain numerical robustness in the Kalman Filter algorithm and reflect a high degree of confidence in the hardware readings, the measurement noise covariance matrix was set to:

$$R_e = I_2 \cdot (\text{deg2rad}(0.1))^2 \tag{7}$$

This value corresponds to a very small standard deviation of 0.1°, prioritizing the sensor data over the model's predictions. To optimize the estimator's transient response, the process noise covariance matrix, Q_e, was tuned through empirical testing on the hardware.

Figure [X] presents a comparative analysis of the LQE performance for three different weights: $Q_e = 0.1 \cdot I_5$, $Q_e = I_5$ and $Q_e = 10 \cdot I_5$. By examining the plots of the measured angles (α, β) against their respective estimates $(\hat{\alpha}, \hat{\beta})$, it is evident that the configuration with $Q_e = 10 \cdot I_5$ provided superior tracking fidelity. This setup allowed the estimates to converge rapidly to the physical values with minimal lag, ensuring that the velocity states (which are not directly measured) were also accurately derived. Based on this visual validation, the more aggressive tuning was selected for the subsequent stages of the project.

2) LQG Testing:

3) Analysis and Problems:

B. Propused Solutions and Simulation

C. Final Hardware Testing and Validation

IV. Conclusion

V. Anexos